C PROGRAMMING — ASSIGNMENT 1

Task 6: Read and Print Data Correctly

Marks: 4 | Total assignment: 50 marks

Format specifiers

Use %d for int, %u for unsigned int, %ld for long, %lld for long long, %f for float in printf, and %lf for double in printf (for scanf, %f expects float* and %lf expects double*). Use %c for a character and %s for a character array. Match each conversion specifier to the exact pointer/value type; a mismatch can cause undefined behaviour. Check scanf's return value: it should equal the number of successfully assigned fields. For robust recovery, fgets followed by strtol can detect malformed or out-of-range input and discard the rest of a line.

Paise formatting

For non-negative total paise p, print Rs p/100 and p%100 using printf("Rs %lld.%02lld", p/100, p%100). For negative values, handle the sign separately; a correct billing system rejects negative bills.

Complexity

Reading a line of length L and parsing it takes O(L) time and O(L) buffer space (or O(B) for a fixed B-byte buffer). scanf can be concise but invalid input may remain unread and cause repeated failures unless the offending input is consumed. fgets provides line boundaries and better recovery control.

C Source Code / Implementation

```
#include <stdio.h>
#include <stdlib.h>
#include <errno.h>
#include <limits.h>

int main(void) {
    char line[128];
    long previous, current;
    char *end;
    printf("Enter previous and current readings (0..65535): ");
    if (!fgets(line, sizeof line, stdin)) {
        fprintf(stderr, "Input error\n"); return 1;
    }
    errno = 0;
    previous = strtol(line, &end, 10);
    if (end == line || errno || previous < 0 || previous > 65535) {
        fprintf(stderr, "Invalid previous reading\n"); return 1;
    }
    while (*end == ' ' || *end == '\t') ++end;
    if (*end == '\n' || *end == '\\0') {
        if (!fgets(line, sizeof line, stdin)) {
            fprintf(stderr, "Missing current reading\n"); return 1;
        }
        end = line;
    }
    errno = 0;
    current = strtol(end, &end, 10);
    if (errno || current < 0 || current > 65535) {
        fprintf(stderr, "Invalid current reading\n"); return 1;
    }
    unsigned int units = (unsigned short)((unsigned short)current - (unsigned short)previous);
    long long bill = 5000LL;
    printf("Units: %u\nBill: Rs %lld.%02lld\n", units, bill / 100, bill % 100);
    return 0;
}
```

Submission note

Compile and run the code on your own system, save actual terminal output/screenshots, and include them in your report. Review the code and be prepared to explain it during the viva. Any machine-specific measurements must come from your own execution.